

Modeling Interactions & Behavior Revision

Student Study Edition · Golden Master design · 12 structured lessons with original course visuals, examples, activities, and study aids.

01 · Smart Parcel Locker Pickup — One Scenario, Five UML Views

02 · Use Case View — Who Wants What?

03 · Class View — What Exists and Relates?

04 · Sequence View — Who Interacts, and When?

05 · State View — What Changes Over Time?

06 · Activity View — How Does Work Flow?

TOPIC 10 · STUDY MAP

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TOPIC 10 · LESSON 01 · 1 OF 12 LESSONS

01 · ANCHOR SCENARIO

Smart Parcel Locker Pickup — One Scenario, Five UML Views

Begin with one complete scenario. Every model in the next pages is derived from this same source of truth.

Practice Lab · Topic 10 Mission 10 — Make Five Models Tell One Story

Review one smart-parking scenario across Use Case, Class, Sequence, State, and Activity views. Trace requirements, detect contradictions, choose the right UML view, and approve only a coherent model set.

ANCHOR SCENARIO · SMART PARCEL LOCKER PICKUP SYSTEM

The scenario we will model throughout Topic 10



Smart Parcel Locker Pickup — the single anchor scenario used throughout Topic 10.

READ THE SCENARIO ONCE · REUSE IT IN EVERY MODEL

An online retailer delivers customer parcels to a network of smart parcel lockers. When a courier arrives at a locker location, the courier identifies the parcel and the system assigns an available compartment that can accommodate it. The courier places the parcel inside, closes the door, and the system records that the parcel is ready for pickup. The customer is then notified through the mobile app and receives a unique pickup code with an expiration time. Later, the customer arrives at the locker and enters the pickup code. The smart locker sends the code to the pickup service for validation. If the code is valid and has not expired, the correct compartment is unlocked and the customer removes the parcel. The locker detects that the parcel has been collected, the pickup service records the successful pickup, and the mobile app shows a confirmation. If the code is invalid, the customer may retry while attempts remain. If the code has expired, pickup is blocked and the customer is instructed to request a new code. The system must never open a compartment for an invalid or expired code, and a successful pickup can be recorded only after the parcel has actually been removed.

We will model the same scenario through five complementary UML viewpoints.



Use Case Diagram

Who uses the system, and what goals do they achieve?



Class Diagram

What concepts exist, what do they know, and how are they related?



Sequence Diagram

Which participants exchange messages, and in what order?



State Diagram

How does one tracked subject change through its lifecycle?

**Activity Diagram**

How does work flow across actions, decisions, and responsibilities?

MODELING RULE

Derive every model from the same requirements. Each view answers a different question, but all five must tell one consistent system story.

TOPIC 10 · LESSON 02 · 2 OF 12 LESSONS

02 · USER-GOAL LENS

Use Case View — Who Wants What?

Review the system boundary, actors, goals, and include/extend relationships using the Smart Parcel Locker scenario.

**Question answered by this view**

Who interacts with the Smart Parcel Locker Pickup System, and which user goals does the system support?

Important distinction

The Use Case Diagram provides the **goal and scenario context**. It is the starting view for the detailed modeling that follows; Class adds structure, while Sequence, State, and Activity describe dynamic behavior.

Primary actors

Customer · Courier

System boundary

Smart Parcel Locker Pickup System

Main goals

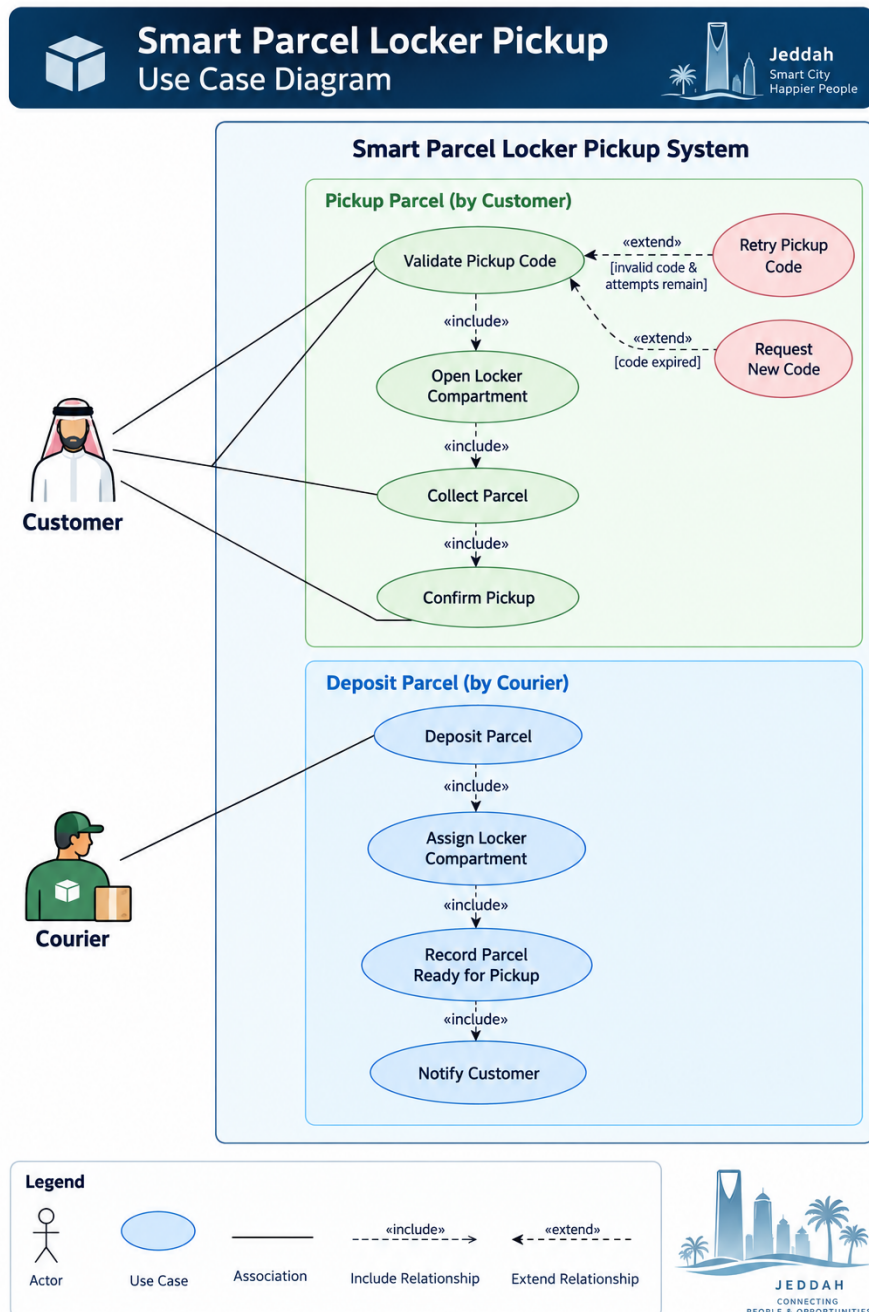
Deposit parcel · Collect parcel

Supporting behavior

Assign locker · Validate code · Notify ·
Confirm pickup

What we intentionally model

The courier wants to **deposit a parcel**; the customer wants to **collect a parcel**. Internal services such as assigning a compartment, validating a code, opening the locker, and confirming pickup support those user goals. They are not separate actors when they are inside the system boundary.



Model solution: actors, goals, system boundary, and supporting use-case relationships.

Review cue: Use Case Diagram = **actors + user goals + system boundary**. It does not show message order, object structure, or state changes.

TOPIC 10 · LESSON 03 · 3 OF 12 LESSONS

03 · STRUCTURE LENS

Class View — What Exists and Relates?

Before following messages, states, and workflows, identify the domain concepts that those behavioral views operate on.



Question answered by this view

What important concepts exist in the Smart Parcel Locker domain, what information do they own, and how are they related?

Core entities

Customer · Courier · Parcel · SmartLocker · Compartment

Control concepts

PickupCode · PickupService · MobileApp

Key relationships

Locker contains compartments · Customer owns parcel · Parcel has pickup authorization

Why it matters later

Sequence lifelines, state subjects, and activity objects should reuse this vocabulary.

Where the Class Diagram fits

Use Case

What goals must the system support?



Class / Domain Model

What concepts exist and relate?



Sequence

Which instances collaborate?



State

Which object changes?



Activity

How does work flow?

Candidate classes from the scenario

Parcel

Tracks identity, size, readiness, and pickup status.

SmartLocker

Provides physical access and owns compartments.

Compartment

Has size, availability, and occupancy state.

PickupCode

Has value, expiration, and remaining attempts.

Customer

Receives notification and collects the parcel.

Courier

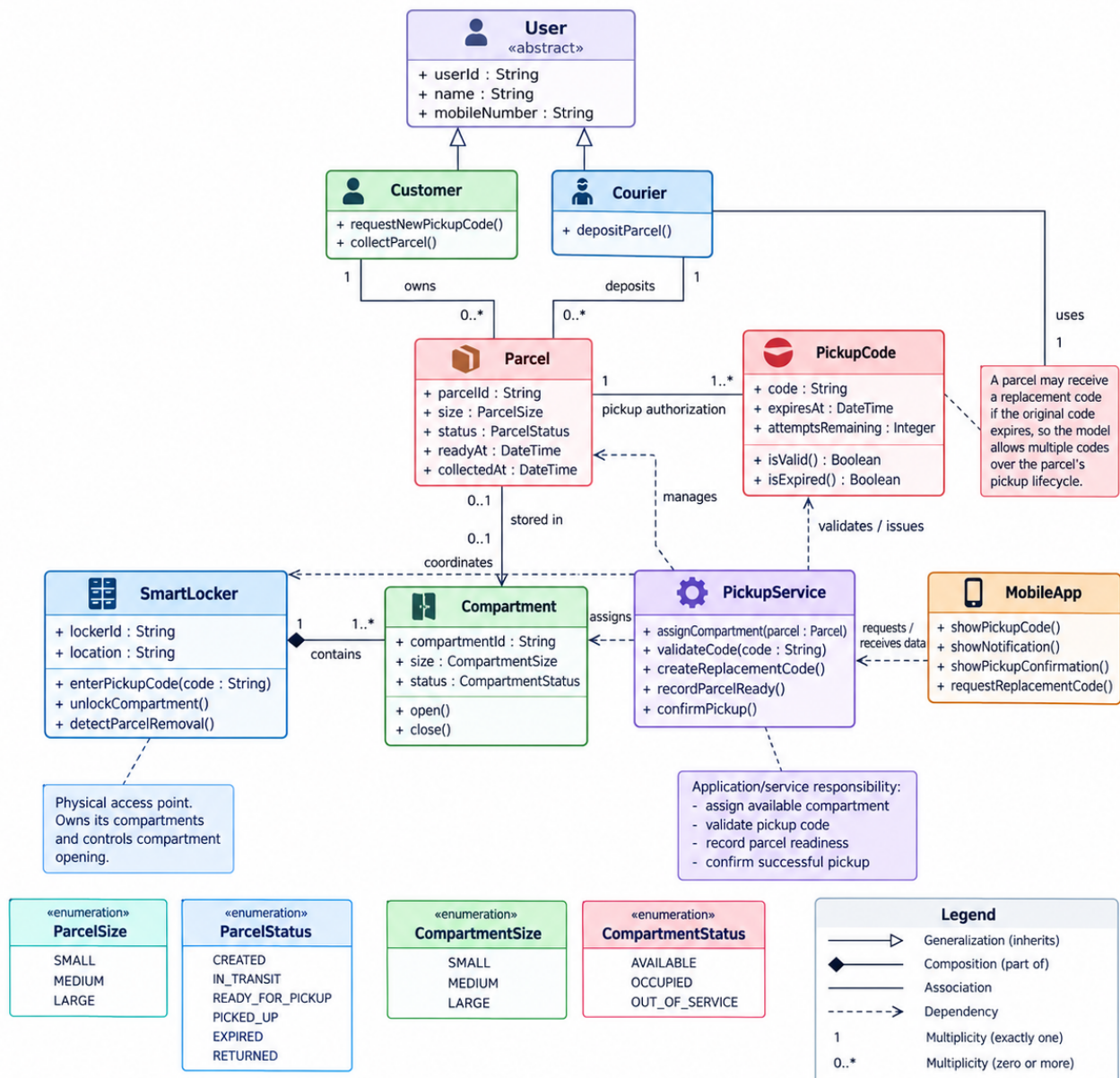
Deposits the parcel into an assigned compartment.

PickupService

Validates codes and records pickup completion.

Smart Parcel Locker Pickup System

UML Class Diagram — Domain / Structural View



Model solution: structural view of the same Smart Parcel Locker Pickup scenario.

Review cue: Class Diagram = **classes + attributes + operations + associations + multiplicities + structural relationships**. It explains what exists, not the temporal order of behavior.

INTEGRATION RULE

Behavioral diagrams should reuse the same domain vocabulary. If a Sequence Diagram introduces a participant or a State Diagram tracks an object that has no conceptual home in the model, investigate the inconsistency.

TOPIC 10 · LESSON 04 · 4 OF 12 LESSONS

04 · INTERACTION LENS

Sequence View — Who Interacts, and When?

Zoom into the Collect Parcel use case and review ordered messages among the customer, locker, pickup service, and mobile app.

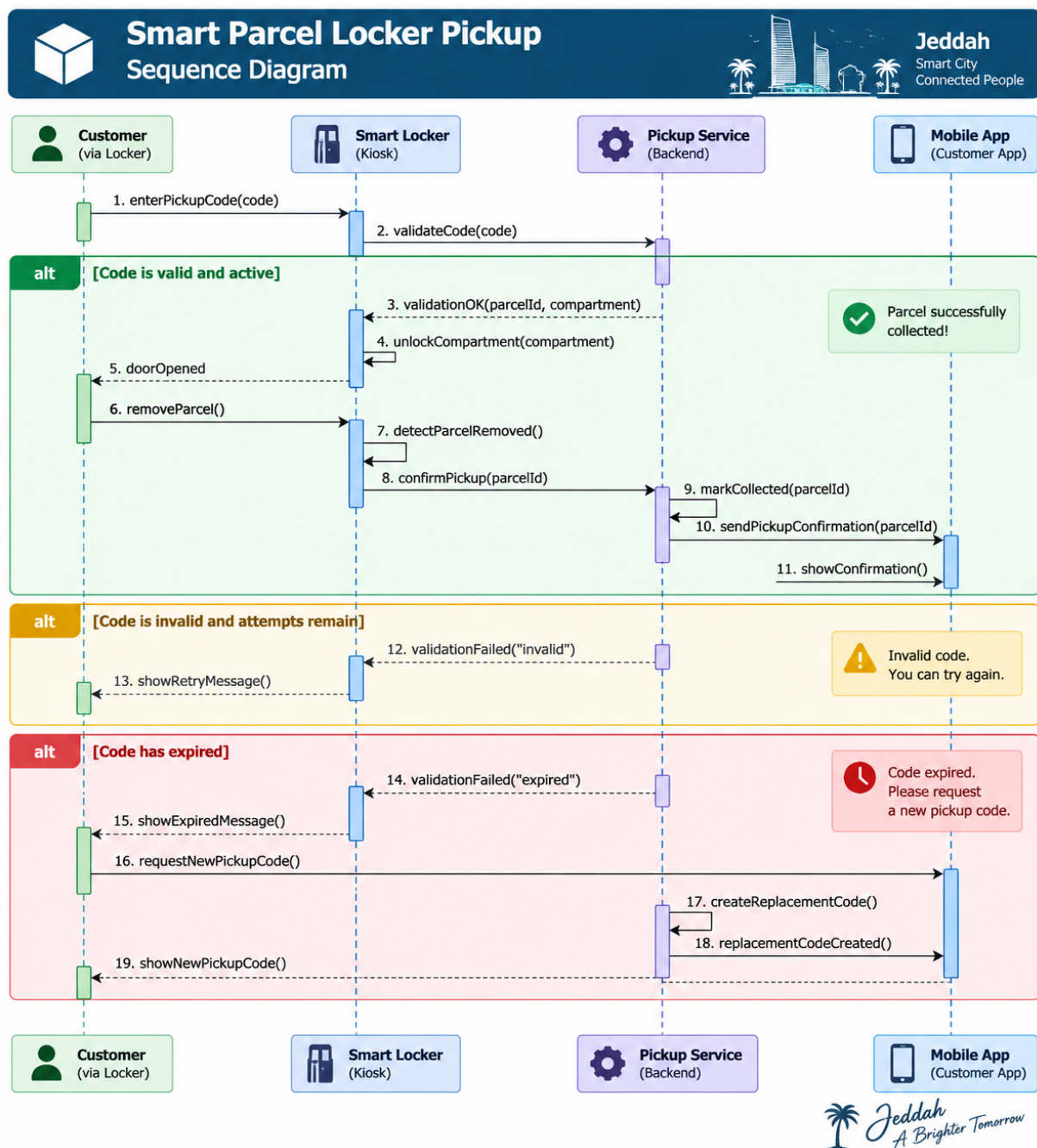
**Scenario slice**

This page models the successful and unsuccessful message exchanges that occur when a customer tries to collect a parcel.



What this view adds

Unlike the Use Case and Class views, the Sequence Diagram reveals **who sends each message, when validation occurs, and how interaction alternatives are handled**. The participants should be traceable to actors or domain/system concepts already identified.



Model solution: ordered interactions for the Collect Parcel scenario.

Review cue: Sequence Diagram = **participants + lifelines + ordered messages + interaction alternatives.**

TOPIC 10 · LESSON 05 · 5 OF 12 LESSONS

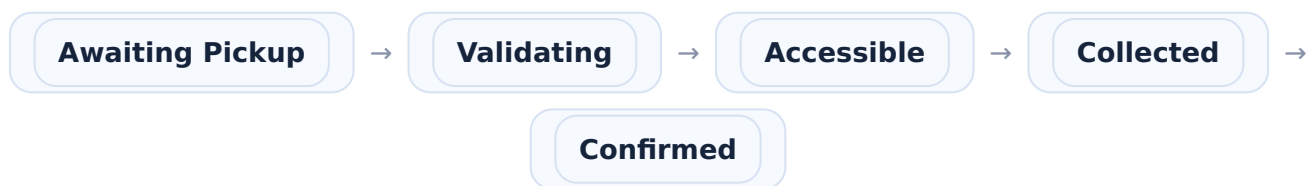
05 · LIFECYCLE LENS

State View — What Changes Over Time?

Track one subject—the Parcel Pickup—and review how events and guards move it between meaningful states.

**Tracked subject: Parcel Pickup**

Do not model every action as a state. Keep only conditions that persist long enough to matter to the lifecycle.



What this view adds

The State Diagram ignores detailed message collaboration and focuses on the changing condition of one tracked subject. Events such as **code entered** and **parcel removed** trigger transitions; guards distinguish valid, invalid, retry, blocked, and expired outcomes.

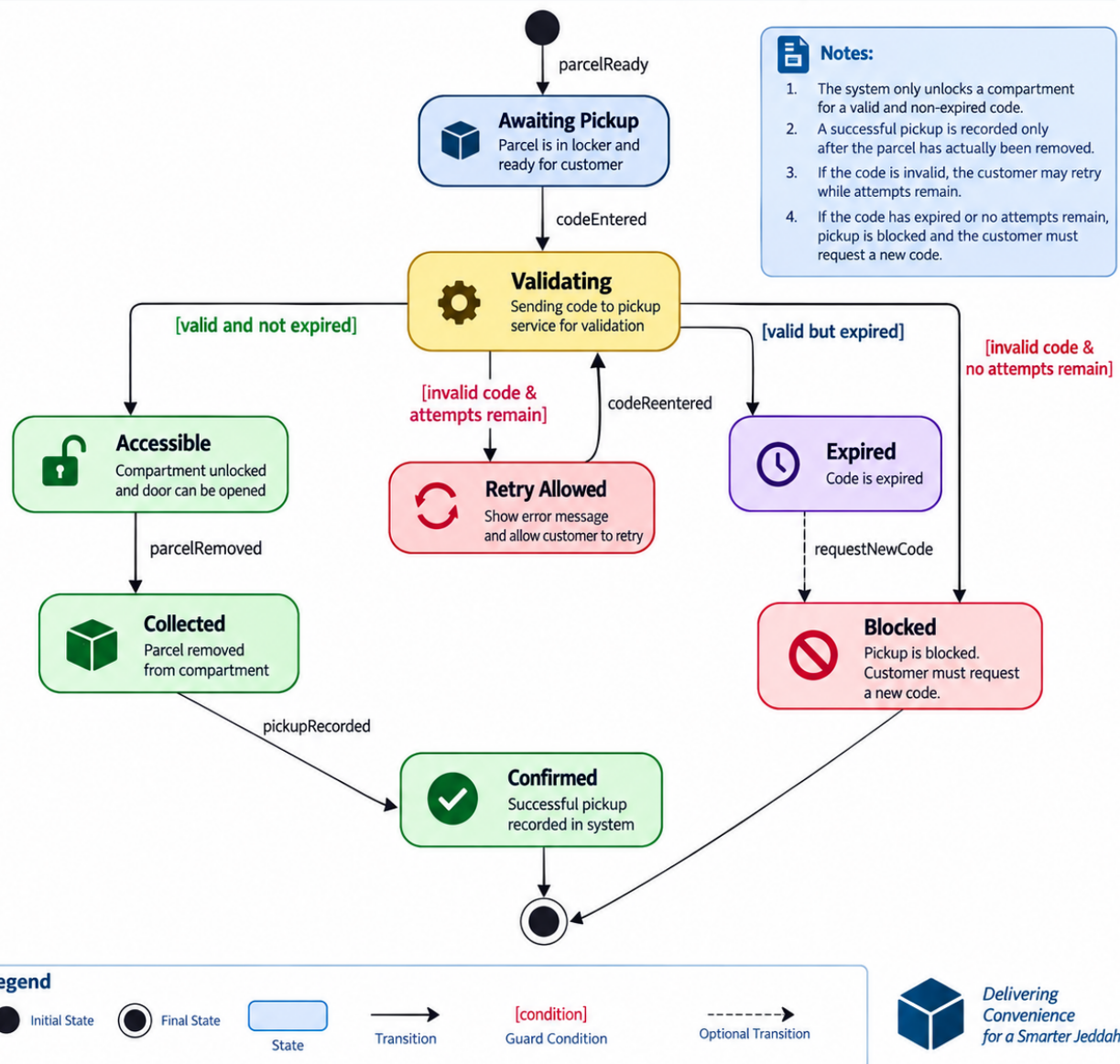


Smart Parcel Locker Pickup State Diagram



Jeddah
Smart City
Connected People

Jeddah
Always Ahead



Model solution: lifecycle, events, guards, retry behavior, and terminal outcomes.

Review cue: State Diagram = **one subject + states + events + guards + transitions + lifecycle outcomes.**

TOPIC 10 · LESSON 06 · 6 OF 12 LESSONS

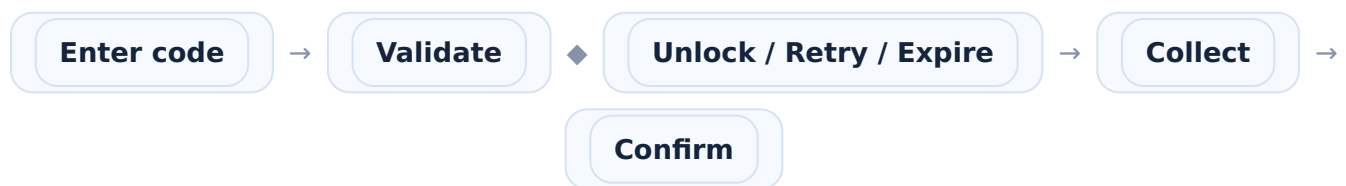
06 · WORKFLOW LENS

Activity View — How Does Work Flow?

Review actions, decisions, swimlane responsibility, and alternative workflow paths for parcel pickup.

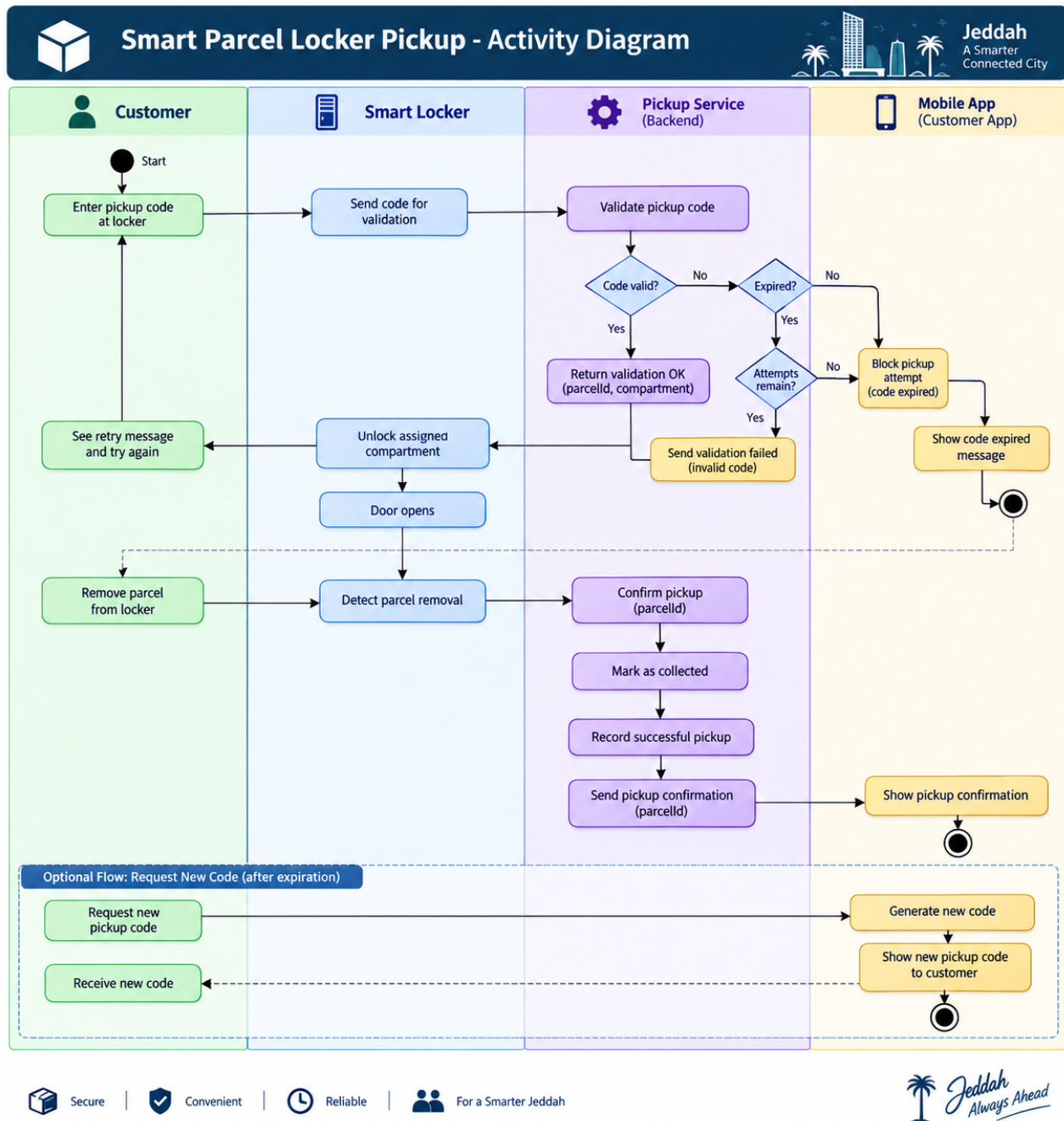
**Workflow question**

What happens from code entry until pickup succeeds, retries, expires, or is blocked—and who performs each action?



What this view adds

The Activity Diagram emphasizes **work and responsibility**. Swimlanes distinguish Customer, Smart Locker, Pickup Service, and Mobile App. Decisions represent alternative paths without turning actions into object states.



Model solution: workflow, decisions, and responsibilities during parcel pickup.

Review cue: Activity Diagram = **actions + control flow + decisions/guards + concurrency when needed + responsibility.**

TOPIC 10 · LESSON 07 · 7 OF 12 LESSONS

07 · CONNECT THE VIEWS

Same Scenario, Five Complementary Models

Connect goals, structure, interactions, lifecycle, and workflow without duplicating the same information in every model.

 Use Case

Reveals: actors, goals, system boundary.

Key fact: Customer collects a parcel.

 Class

Reveals: concepts, attributes, relationships, multiplicities.

Key fact: Parcel, PickupCode, Locker, and Compartment share one structural vocabulary.

 Sequence

Reveals: ordered message collaboration.

Key fact: validation happens before unlock.

 State

Reveals: lifecycle, events, guards.

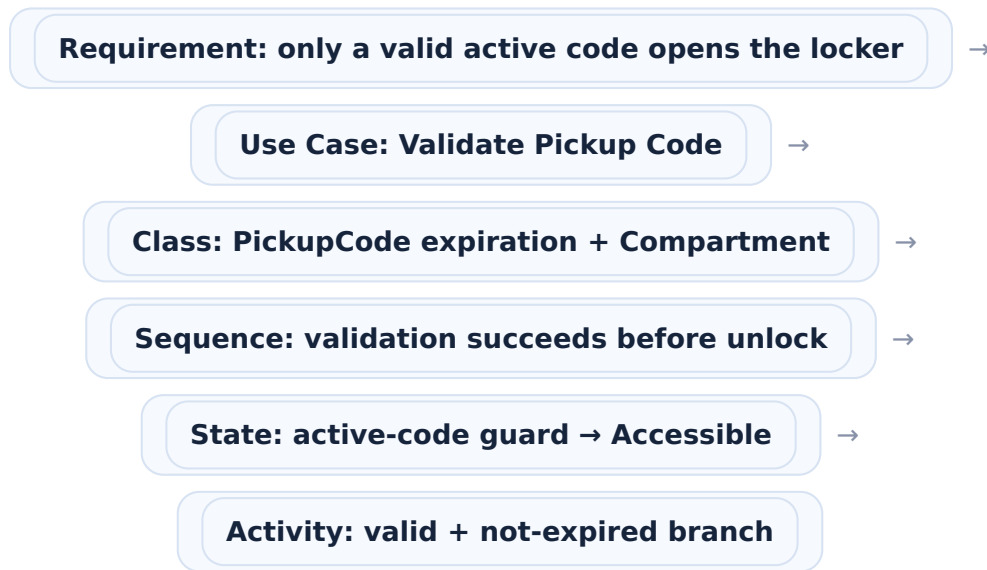
Key fact: a valid active code leads to Accessible.

 Activity

Reveals: workflow, decisions, responsibility.

Key fact: valid, invalid, and expired paths branch differently.

Trace one requirement across five views



Different abstraction levels

Use Case and Class provide goal/context and structure. Sequence, State, and Activity provide detailed behavioral views. They are complementary—not five copies of the same picture.

Consistency does not mean duplication. Each diagram keeps only the information needed by its viewpoint.

TOPIC 10 · LESSON 08 · 8 OF 12 LESSONS

08 · CROSS-MODEL REASONING

Consistency Across Models

Use all five views as mutual checks. Contradictions often reveal missing, ambiguous, or misunderstood requirements.

1

Check goals

Use cases must match behavior implemented in the other views.

2

Check structure

Classes and relationships should support the objects, data, and responsibilities used later.

3

Check messages

Lifecycle-changing messages should agree with state transitions.

4

Check guards

Important conditions should remain compatible across State and Activity views.

5

Check outcomes

Success, retry, expiry, blocking, and completion must mean the same thing everywhere.

Spot the contradiction

Suppose the Sequence Diagram says an **expired code causes the compartment to unlock**, while the State Diagram terminates pickup as **Expired** and the Activity Diagram blocks the path. These models cannot all satisfy the same requirement. Fix the requirement interpretation before polishing any diagram.

Traceability questions

Use Case → Class

Do goals rely on concepts that exist in the domain model?

Class → Sequence

Can key lifelines be traced to actors, objects, or services?

Sequence → State

Do lifecycle-changing messages correspond to valid transitions?

State ↔ Activity

Do guards and outcomes tell the same story?

ENGINEERING HABIT

Use disagreements between models as evidence. They expose missing, inconsistent, or misunderstood requirements.

TOPIC 10 · LESSON 09 · 9 OF 12 LESSONS

09 · TRANSFER THE METHOD

Integrated Case Study

Transfer the five-view method to a new system after mastering the Smart Parcel Locker example.

TRANSFER CASE · EV CHARGING SESSION**Plug in → authorize → charge → stop → pay**

A driver connects an EV to a charging station. The station authorizes the account, starts charging if approved, stops when requested or when charging completes, calculates the cost, and records payment.

 Use Case candidate

Start charging session · Stop charging · Pay for session.

 Class candidate

Driver · ChargingStation · ChargingSession · Connector · Account · Payment.

 Sequence candidate

Driver ↔ Station ↔ Authorization Service ↔ Payment Service.

 State candidate

Idle → Authorizing → Charging → Completing → Paid.

 Activity candidate

Connect → authorize → [approved?] → charge → calculate → pay.

TRANSFER PRINCIPLE

The five questions stay stable even when the domain changes. Start from one scenario and derive only the views needed to answer those questions.

TOPIC 10 · LESSON 10 · 10 OF 12 LESSONS

10 · CHOOSE THE VIEW

Model Selection Challenges

Choose among Use Case, Class, Sequence, State, and Activity by first identifying the modeling question.

 “Show what a Rider and Driver want from the system.”

Best lens: **Use Case**

 “Show Ride, Rider, Driver, Vehicle, and their relationships.”

Best lens: **Class**

 “Show how app, API and payment gateway exchange calls.”

Best lens: **Sequence**

 “Show how an Order changes from Created to Delivered or Cancelled.”

Best lens: **State**

 “Show checkout steps, decisions, and parallel checks.”

Best lens: **Activity**

Sometimes one question needs multiple views

“When payment succeeds, the order becomes Paid and fulfillment begins” touches multiple viewpoints: **Class** identifies Order and Payment, **Sequence** can show payment messages, **State** captures the lifecycle change, and **Activity** shows workflow continuation. Add **Use Case** when the user goal or system boundary is part of the question.

Fast rule: ask the modeling question first. Choosing a diagram because it is familiar is not a modeling decision.

TOPIC 10 · LESSON 11 · 11 OF 12 LESSONS

11 · PRACTICE STUDIO

Integrated Modeling Studio

Build a coherent five-model set from one scenario. Do not start by drawing—first decide what each view must answer.

Studio challenge · Ride-hailing pickup

Rider requests a trip. The platform searches for a driver. A driver may accept or the request may time out. After acceptance, the driver travels to pickup, the rider boards, and the trip begins.

1

Use Case

Identify actors, goals, and boundary.

2

Class

Extract domain concepts, key attributes, and relationships.

3

Sequence

Choose lifelines and key messages for one scenario slice.

4

State

Choose one tracked subject and meaningful lifecycle states.

5

Activity

Show actions, decisions, handoffs, and concurrency if present.

Cross-check after drawing



SUBMISSION TEST

Can you explain why each element belongs in that diagram instead of another one—and can you trace the same requirement across views?

TOPIC 10 · LESSON 12 · 12 OF 12 LESSONS

12 · MASTERY CHECK

Integrated Review & Exam Readiness

Confirm that you can select, read, connect, and critique all five UML views used in this integrated review.

Use Case

Find actors, goals, boundary, include/extend

Class

Read concepts, attributes, relationships, multiplicities

Sequence

Read lifelines, messages, order, alternatives

State

Trace states, events, guards, transitions

Activity

Follow actions, branches, concurrency, swimlanes

Integration

Detect contradictions and justify model selection

60-second model-selection rule

WHO + GOAL?

Use Case

WHAT EXISTS + RELATES?

Class

WHO + MESSAGE ORDER?

Sequence

ONE SUBJECT + LIFECYCLE?

State

WORK + BRANCHING?

Activity

Final self-check

✓ I can derive five complementary UML views from one shared scenario.

✓ I can explain what each view includes and intentionally omits.

✓ I can distinguish structural modeling from detailed behavioral modeling.

✓ I can detect contradictions across goals, structure, messages, states, and workflows.

- ✓ I can justify my diagram choice from the modeling question.

TAKEAWAY

Modeling is not drawing five versions of the same picture. It is viewing one system through purposeful abstractions while keeping the underlying story and terminology consistent.